

KARIM HASHAD

Unity Developer | VR / AR & Multiplayer Specialist | Backend (.NET) Developer

Cairo, Egypt | kareememad612005@gmail.com | +20 10 3239 2337 | [LinkedIn](#) | [GitHub](#) | [Portfolio / Itch.io](#)

PROFESSIONAL SUMMARY

Unity Developer with 2+ years of hands-on experience shipping immersive VR/AR applications, including a published Google Play Store title. Skilled in building real-time multiplayer experiences with Photon PUN2, optimizing asset pipelines with the Addressable Asset System, and delivering performant cross-platform builds for Meta Quest 2/3 and Android ARCore devices. Backend-capable through ASP.NET Core and Entity Framework, able to build the full client-server stack behind a game (matchmaking, leaderboards, accounts) rather than relying solely on third-party services. B.Sc. in Computer Science and Artificial Intelligence (Benha University, 2024). Looking to bring strong C#, performance-optimization, and XR-interaction expertise to a Unity Developer, Gameplay Programmer, or XR/VR Developer role on a multiplayer or immersive-tech team.

TECHNICAL SKILLS

Programming Languages: C#, C++, JavaScript, SQL, HTML, CSS

Game Engines & Core Unity: Unity Engine (2022 LTS+), Unity UI, TextMeshPro, Unity Profiler, Scriptable Objects, Object Pooling, Addressable Asset System

VR / XR Technologies: XR Interaction Toolkit, Meta Quest 2/3 (native + Link), AR Foundation, Google ARCore, Hand Tracking, Locomotion & Teleportation Systems

Multiplayer Networking: Photon PUN2, Photon Fusion (working knowledge), Unity Netcode for GameObjects, Client-Server Synchronization, State Replication, Matchmaking Concepts

Backend & Cloud / APIs: ASP.NET Core, Entity Framework Core, RESTful API Design, JWT Authentication, SQL Server, Google Play Console APIs, Postman

Architecture & Patterns: OOP, SOLID Principles, Clean Architecture, Design Patterns (Singleton, Observer, Factory, Object Pool), MVC

Tools & Platforms: Git, GitHub, Android Studio (build/deploy), Visual Studio, Performance Profiling, Cross-Platform Deployment (Android, Quest)

PROFESSIONAL EXPERIENCE

Unity Developer — VR/AR Specialist

Freelance / Contract Projects • 2022 – Present (2+ years)

- Built and shipped multiple VR and AR applications end-to-end in Unity, owning interaction design, performance budgets, and platform-specific optimization for mobile and standalone headsets.
- Engineered an Addressable Asset System pipeline across projects, cutting initial build size and enabling on-demand asset streaming to reduce load times on memory-constrained mobile and headset hardware.
- Implemented real-time multiplayer with Photon PUN2, synchronizing player transforms, object ownership, and shared-state events across concurrent sessions with minimal perceived latency.
- Took an AR application through the full lifecycle — prototyping, plane-detection/object-placement features, QA, and store compliance — to a successful public release on Google Play Store.
- Built core VR interaction systems (grab, throw, teleport-locomotion, hand-tracked UI) using XR Interaction Toolkit, tuned for comfort and reduced motion sickness.
- Profiled and optimized rendering, physics, and draw calls for Meta Quest 2/3 and Android AR devices using Unity Profiler, sustaining target frame rates (72+ FPS on Quest) for comfortable XR use.
- Partnered directly with designers and stakeholders to translate product requirements into technical specs, maintaining a clean, modular, SOLID-compliant codebase for long-term maintainability.

KEY PROJECTS

AR Application — Published on Google Play Store

Unity AR Developer • Unity, AR Foundation, Addressables, Android

- Challenge: deliver stable real-world object placement and persistent AR content on a wide range of low-to-mid-tier Android devices without sacrificing frame rate.
- Contribution: implemented plane detection and object-placement interactions with AR Foundation; integrated the Addressable system for on-demand content loading to keep the install size lean.
- Result: shipped a stable, store-approved AR app on Google Play with smooth, responsive interactions across tested device tiers.

Interactive VR Multiplayer Experience

Unity VR Developer • Unity, XR Interaction Toolkit, Photon PUN2, Meta Quest 2

- Challenge: support multiple concurrent users interacting with shared, physics-driven objects in the same VR space without desync or performance loss.
- Contribution: built grab/throw/teleport interaction mechanics and hand-tracked UI; implemented Photon-based session sync for shared object state and player presence across clients.
- Result: maintained 72+ FPS on Meta Quest 2 in multiplayer sessions, preserving comfort and responsiveness under concurrent network load.

BattleField — Third-Person Action Shooter (In Development)

Unity • C# • Dual-Wield Weapon System • Object Pooling • Fire Rate Management • VFX Pipeline • Collision Response • Third-Person Controller → [Full breakdown & demo on GitHub](#)

Kitchen Chaos — Single-Player Cooking / Time-Management Game

Unity • C# • State Machines • ScriptableObjects • Interaction System • Recipe & Scoring System • UI / Audio Systems → [Full breakdown & demo on GitHub](#)

Hotel Reservation Web Application

Backend Developer • ASP.NET Core, Entity Framework Core, SQL Server, JWT

- Challenge: provide a secure, well-structured backend for search, booking, and account management that could support a real-time client (relevant to game backend/services work).
- Contribution: designed the SQL Server schema and Entity Framework Core data layer; built RESTful endpoints with JWT authentication and role-based authorization.
- Result: delivered a fully tested API suite (validated in Postman) demonstrating the same service-layer skills needed for game backends — accounts, persistence, and secure session handling.

EDUCATION

B.Sc. Computer Science and Artificial Intelligence

Faculty of Computer Science and Artificial Intelligence, Benha University, Egypt • Graduated 2024

- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Graphics, Artificial Intelligence, Database Systems, Software Engineering.

CERTIFICATIONS

- Add relevant certifications here once completed — see recommendations below (e.g., Unity Certified Programmer, Unity XR / VR Development course).

LEADERSHIP & VOLUNTEERING

Volunteer Technical Instructor — IEEE Student Branch

- Designed and delivered hands-on workshops on embedded systems and C fundamentals to student cohorts, translating professional IOT systems workflows into beginner-friendly material.
- Designed and delivered hands-on workshops on Embedded Systems and C programming fundamentals, translating industry-standard hardware and software concepts into beginner-friendly learning experiences.